```
log using "C:\Users\Aditi\Downloads\NEW2.smcl"
import excel "C:\Users\Aditi\OneDrive\Desktop\AMM_Final\AMM_Data.xlsx",
sheet("AMM_Data") firstrow
describe
summarize
gen id = _n
gen pop_density = population_total / area_m2
gen rent_income_ratio = median_rent_2010_adj / median_income_2010_adj
gen renter_proxy = 1 - pct_owner_occupied
*STEP 4: CREATE GENTRIFICATION INDEX
egen z_income = std(median_income_2010_adj)
egen z_education = std(pct_bach)
egen z_rent = std(median_rent_2010_adj)
*Create index
gen gentrification = z_income + z_education + z_rent
count if !missing(gentrification)
*STEP 5: BASELINE ELIGIBILITY
*Create baseline income (1980)
gen base_income = median_income_2010_adj if year==1980
sort year
replace base_income = base_income[_n-1] if missing(base_income)
replace base_income = base_income[_n-1] if missing(base_income)
replace base_income = base_income[_n-1] if missing(base_income)
*Create eligible dummy
sum base_income if year==1970
gen eligible = (base_income < r(p50))
*STEP 6: RUN MAIN MODEL
reg median_income_2010_adj gentrification pop_density pct_unemployed pct_assisted_income
pct_owner_occupied i.year, cluster(year)
*STEP 7: POVERTY MODEL
reg pct_poverty gentrification pop_density pct_unemployed pct_assisted_income i.year,
cluster(year)
*STEP 8: UNEMPLOYMENT MODEL
reg pct_unemployed gentrification pop_density pct_assisted_income i.year, cluster(year)
*STEP 9: REALLOCATION (SKILL CHANGE)
reg pct_bach gentrification pop_density i.year, cluster(year)
*STEP 10: DISTRIBUTIONAL EFFECTS
gen interaction = gentrification * renter_proxy
reg median_income_2010_adj gentrification interaction pop_density pct_unemployed i.year,
cluster(year)
*STEP 11: CHILD POVERTY (OPPORTUNITY)
reg pct_poverty_under_18 gentrification pop_density pct_unemployed i.year, cluster(year)
*STEP 12: ROBUSTNESS (ONLY ELIGIBLE TRACTS)
```

```
reg median_income_2010_adj gentrification pop_density pct_unemployed if eligible==1,
cluster(year)
reg population_total gentrification pop_density i.year, cluster(year)
*H5 HYPOTHESIS TO BE MORE STRONG
reg median_rent_2010_adj gentrification pop_density i.year, cluster(year)
reg rent_income_ratio gentrification pop_density i.year, cluster(year)
*H6 HYPOTHESIS TO BE MORE STRONG
reg pct_hs_dropout gentrification pop_density i.year, cluster(year)
reg population_total gentrification pop_density i.year, cluster(year)



*IV Strategy
gen initial_bach = pct_bach if year==1980
gen initial_density = pop_density if year==1980
gen initial_vacancy = pct_vacant_housing if year==1980

sort year
replace initial_bach = initial_bach[_n-1] if missing(initial_bach)
replace initial_density = initial_density[_n-1] if missing(initial_density)
replace initial_vacancy = initial_vacancy[_n-1] if missing(initial_vacancy)

bysort year: egen metro_bach_trend = mean(pct_bach)
bysort year: egen metro_rent_trend = mean(median_rent_2010_adj)

gen iv_skill = initial_bach * metro_bach_trend
gen iv_density = initial_density * metro_rent_trend

ivregress 2sls median_income_2010_adj (gentrification = iv_skill iv_density) pop_density
pct_unemployed pct_assisted_income i.year, cluster(year)

bysort year: egen avg_gentrification = mean(gentrification)

gen gentrification_spill = avg_gentrification - gentrification

*DYNAMIC MODEL
gen gentrification_lag1 = gentrification[_n-1]
gen gentrification_lag2 = gentrification[_n-2]
reg median_income_2010_adj gentrification gentrification_lag1 gentrification_lag2 pop_density
pct_unemployed pct_assisted_income i.year, cluster(year)

*HETEROGENEITY ANALYSIS
gen interaction_vacancy = gentrification * pct_vacant_housing
gen interaction_owner = gentrification * pct_owner_occupied
```

```
gen interaction_assist = gentrification * pct_assisted_in
reg pct_poverty gentrification interaction_assist interaction_vacancy pop_density
pct_unemployed i.year, cluster(year)
```